

Theory:

1. What are Python modules and packages? How do you import them?
2. What is the difference between shallow copy and deep copy?
3. What is exception propagation in Python?
4. What are lambda functions? Provide examples.
5. Name any 3 useful Python standard libraries and their use.

Coding:

1. Write a Python program using the math module to calculate area and circumference of a circle.
2. Handle multiple exceptions (e.g., ZeroDivisionError and ValueError) in a single try-except block.
3. Write a program that merges two dictionaries and removes duplicate keys.

```
1 dict1 = {'a': 1, 'b': 2, 'c': 3}
2 dict2 = {'b': 4, 'd': 5}
3
4 merged = **dict1, **dict2}
5 print("Merged dictionary:", merged)
```

Merged dictionary: {'a': 1, 'b': 4, 'c': 3, 'd': 5}

=== Code Execution Successful ===

```
1- try:
2     num = int(input("Enter a number: "))
3     result = 100 / num
4     print(f"Result: {result}")
5- except (ZeroDivisionError, ValueError) as e:
6     print(f"Error occurred: {e}")
```

Enter a number: 20
Result: 5.0
=== Code Execution Successful ===

```
import math

radius = float(input("Enter the radius of the circle: "))
area = math.pi * radius ** 2
circumference = 2 * math.pi * radius

print(f"Area: {area:.2f}")
print(f"Circumference: {circumference:.2f}")
```

Enter the radius of the circle: 450
Area: 636172.51
Circumference: 2827.43
=== Code Execution Successful ===

Q1Ans>1. Python Modules and Packages

- Modules are Python files (.) containing functions, classes, or variables.
- Packages are directories with a file that group related modules.

Importing:

```
import math
```

```
from datetime import datetime
```

```
import mypackage.mymodule
```

Q2Ans>. Shallow Copy vs Deep Copy

- Shallow Copy: Copies the outer object but not nested objects. Changes to nested items affect both copies.
- Deep Copy: Recursively copies all nested objects, creating a fully independent clone.

Example:

```
import copy
```

```
original = [[1, 2], [3, 4]]
```

```
shallow = copy.copy(original)
```

```
deep = copy.deepcopy(original)
```

Q3Ans>Exception Propagation

- When an exception occurs, Python searches for a matching block in the current function.
- If not found, it propagates up the call stack until handled or crashes the program.

Example:

```
def inner():  
    raise ValueError("Oops!")
```

```
def outer():  
    inner()
```

```
try:  
    outer()  
except ValueError as e:  
    print("Caught:", e)
```

Q4Ans> Lambda Functions

- Anonymous, one-line functions using the `lambda` keyword.
- Syntax: `lambda arguments: expression`

Examples:

```
square = lambda x: x ** 2  
add = lambda a, b: a + b
```

```
print(square(5)) # 25  
print(add(3, 4)) # 7
```

Q5Ans> Useful Python Standard Libraries

5. Name any 3 useful Python standard libraries and their use

1.

- Use: Provides mathematical functions like square root, trigonometry, logarithms, and constants like π and e .

- Example:

```
import math  
print(math.sqrt(25)) # Output: 5.0
```

.

- Use: Handles date and time operations including formatting, arithmetic, and current timestamps.

- Example:

```
from datetime import datetime
```

```
now = datetime.now()
print(now.strftime("%Y-%m-%d %H:%M:%S"))
```

Use: Interacts with the operating system—file paths, environment variables, directory management.

- Example:

```
import os
print(os.getcwd()) # Prints current working directory
```